Phishing URL Detection: A Machine Learning Approach

Manish Katel, Sarah Mirza, Dr. Nick Rahimi

The University of Southern Mississippi

THE UNIVERSITY OF SOUTHERN MISSISSIPPI

Introduction: The Cybersecurity Landscape

859,532

Internet crime complaints filed in 2024

 # 1 Crime: Phishing

Top crime among all internet crimes

- Cyberattacks evolve faster than traditional defenses
- Signature-based phishing detection fail against new/unknown urls
- ML-based Phishing URL Detection are essential for real-time monitoring
- High-dimensional low variance feature spaces and class imbalance are the major challenges

Attackers mimic well-known banks. Credit card agencies and e-commerce sites to steal credentials.

Research Purpose

Goal

Design and evaluate the efficient machine learning pipeline and build effective models to detect Good URLS vs Phishing URLS.

Binary Classification

Good vs. Phishing

Models

Random Forest, Decision Tree
Logistic Regression

Scalable Detection

Adaptable pipeline for unseen phishing urls

URL Structure - What's Inside a URL?

What is a URL?

A URL (Uniform Resource Locator) is the unique, human-readable address used to locate specific resources - such as webpages, images, or documents - on the internet.

```
https://www.example.com:443/blog/article/search?docid=720&hl=en#dayone
```

Scheme	Subdomain	Domain	TLD	Port	Path	Query Sep.	Query String	Fragment
https://	www.	example.	com	:443	/blog/.../search	?	docid=720&hl=en	#dayone

URLs contain words, numbers, dots (.), slashes (/), hyphens (-) and special characters - each signals whether a URL is legitimate or malicious.

Normal vs Phishing URL — Examples & Tokens

Normal / Good URL

```
archive.org/stream/watkinsfamilyofn00watk/
watkinsfamilyofn00watk_djvu.txt
```

Tokens:

the	world	heritage
sites	com	

Phishing / Bad URL

```
nobell.it/70ffb52d.../login.SkyPe.com/en/
cgi-bin/verification/login/...index.php?
cmd=_profile-ach&login_access=1322408526
```

Suspicious Tokens:

login	verification	php
UserID	userid	cmd

Background - Why Machine Learning for Phishing Detection?

Traditional Methods Fail

Blacklisting & content analysis cannot keep up with newly created phishing URLs.

Attackers Adapt Fast

New domains and subdomains are trivially generated, bypassing all static rules.

ML Offers Prediction

Models learn structural URL patterns, generalizing to unseen phishing URLs.

High Accuracy Achieved

Random Forest achieves >83% accuracy on URL structural feature sets.

Key URL Features from Literature

`url_length`
Total character count of the URL

`dot_count`
Number of '.' — more dots = suspicious

`slash_count`
Number of '/' — deep paths flag phishing

`hyphen_count`
Hyphens added to fake brand names

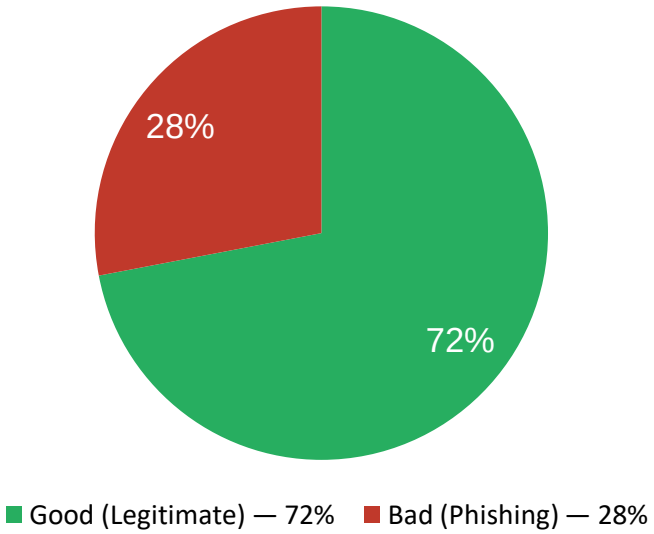
`suspicious_word_count`
Tokens like login, verify, cmd, php

Dataset Description

Phishing Site URL Dataset

Total Entries	549,346 URL samples
Columns	URL (text) + Label (Good / Bad)
Good URLs	72% — Legitimate, benign content
Bad URLs	28% — Phishing / malicious content
Source	kaggle.com/taruntiwarihp/phishing-site-urls

Class Distribution



Methodology

Data Visualization

Word tokenization; exploratory analysis of URL patterns

Data Cleaning

Lowercase conversion, duplicate removal, noise cleaning

Feature Extraction

url_length, dot_count, hyphen count, slash count, suspicious words count

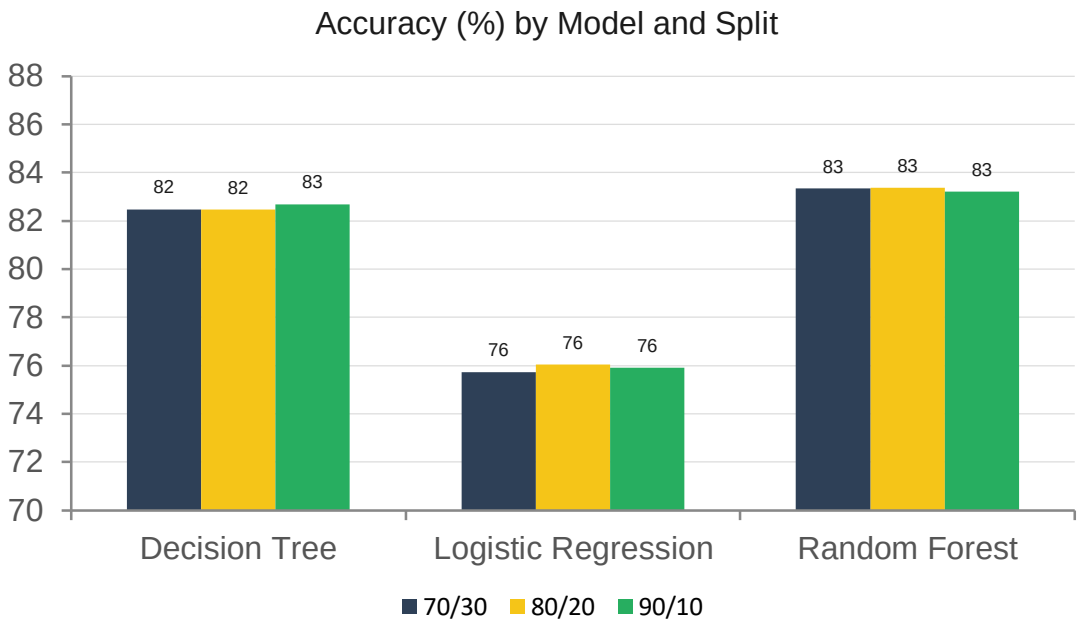
Train / Test Split

Three splits: 70/30, 80/20, 90/10 for robust comparison

Model Training and Evaluation

3 models trained, 10 fold Cross Validation, GridSearchCV and Classification reports

Results - Model Accuracy Comparison



Weighted F1 Score Summary

Model	70/30	80/20	90/10
Decision Tree	83.16	83.14	83.31
Logistic Reg.	76.45	76.72	76.64
Random Forest	**83.92**	**83.96**	**83.82**

Best Model: Random Forest

83.38% Accuracy
83.96% Weighted F1 (80/20 split)

Limitations & Future Directions

Limitations

Imbalanced Dataset

72/28 split causes overrepresentation of 'Good' URL accuracy - models may be biased toward majority.

No Feature Importance

Feature Importance Plot was not generated - unclear which features are most impactful.

Basic Tokenization

Used simple word_tokenize for suspicious words - less context-aware than n-gram methods.

No Real-World Testing

Models were not evaluated on live, externally collected real-world phishing URLs.

Future Directions

Balance the Dataset

Downsample the majority class before training to address class imbalance effectively.

Explainability (SHAP/LIME)

Use SHAP and LIME libraries to visualize and interpret feature contributions.

Better Tokenization

Replace word_tokenize with CountVectorizer using n-gram approach for richer features.

Real-World Validation

Test trained models on newly collected live phishing URLs for deployment-ready performance.

Conclusion

Problem Addressed

Phishing is the #1 internet crime (FBI 2024). Traditional blacklisting and content-analysis methods cannot adapt to newly generated malicious URLs.

Dataset & Features

Trained on 549,346 URLs (72/28 split) using url_length, dot_count, slash_count, hyphen_count and suspicious_word_count as classification features.

Best Model

Random Forest achieved 83.38% accuracy and 83.96% Weighted F1 on the 80/20 split - outperforming Decision Tree and Logistic Regression.

Key Insight

URL structural features are strong discriminators. Phishing URLs are longer with more dots, slashes, and suspicious token patterns.

References

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Thank You

Questions & Discussion

Manish Katel · Sarah Mirza · Dr. Nick Rahimi
The University of Southern Mississippi